

Preparation of capillary for H₂O-NMR

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Tags: NMR AE capillary

Category: Protocol

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Goal

Capillaries with D₂O and H₃PO₄ as reference and lock substance for NMR-measurments in H₂O.

Prerequisites and preparation

D₂O, H₃PO₄

long glas pipettes

lighter

glas cutter

Steps

ca. 1h

Step number	Step description
1: Prepare the solution	In a Vial D ₂ O (1 mL) and concentrated H ₃ PO ₄ (85 %, approx. 5 µL) are added using a 1 mL-syringe. The obtained solution is mixed by shaking.
2: Prepare the glas capillary	A approx. 6 to 7 cm long peace is cut of from thin end of a glas pipette. Therefor the glascutter is set at the desired spot and the glas pipette is held against the glas cutter. The pipette is turned unti the piece breaks of.
3: Fill the capillary	The obtained capillary is held in the obtained soultion. To achieve an sufficient filling high, both the capillary and the GC-Vial containg the solution are tillted. The filling high should be approx. 5 cm.
4: Seal the capillary	Place the liquid in the middel of the capillary, so that on both sides a bit air is present. Then heat one end carfully with a lighter till the glas melts. Thereby rotate the capillary until it is sealed. For good sealing a sphere should form at the end. If water starts to evaporate this can be seen by the building of an bubble in the glas. In that case the capillary will not be sealed, therefor the capillary can be discarded.
5: Seal the other end	If not sufficient air seperates the end of the capillary and the water move air bubbles upwards to the open end by gently (!) snipping against the cappillary. Than seal the capillary with a lighter as in 4.
6: Check the capillary	To check if the capillary is sealed the capillary is placed in a Schlenk tube and is set under vacuum. If leaking occurs, the capillary can be discarded, otherwise the capillary can by used.



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